

Aliaksei Ivanou | Senior C++ Software Engineer

Qt · Embedded Linux · Yocto · SIP/WebRTC · HMI

aliaksei.ivanou.by@icloud.com | +375 29 559 85 56 | [LinkedIn](#) | [GitHub](#)

Relocating to Warsaw | Available on-site from 26 Sep 2026

Polish national D visa valid 26 Sep 2026 - 25 Sep 2027

Senior C++ Software Engineer with 6+ years of commercial C/C++ experience and 13+ years across software engineering and R&D. Specialized in Qt-based HMI, Embedded Linux, Yocto/BitBake, NXP i.MX, and SIP/RTP/WebRTC real-time communication. Hands-on experience designing production UI architecture and delivering multithreaded/asynchronous systems, device integration, networking, Linux debugging, and automated testing, with code reviews, technical mentoring, onboarding, and technical interview support in a 6-person engineering team.

Technical Skills

- **C++:** C++11/14/17, STL, C, OOP, RAII, smart pointers, object lifetime management, multithreading/concurrency
- **Architecture & Engineering Practices:** design patterns, MVC/MVVM-style UI architecture, code reviews
- **Linux:** Embedded Linux, Yocto/BitBake, NXP i.MX, systemd, Linux troubleshooting, service configuration, shell/Bash
- **Qt/HMI:** Qt 5.15, Qt Widgets, Qt Model/View, Qt signals/slots, QObject, QThread/QtConcurrent, queued connections, QSS
- **Networking & Real-Time Media:** TCP/IP, SIP, RTP/SDP, WebRTC, WebSockets, liblinphone, mediasoup, GStreamer, V4L2, SNMP, DHCP, TFTP
- **Build, Debugging & CI/CD:** CMake, GCC, gdb, Valgrind, core dump analysis, Git, GitLab CI, Jenkins, SSH
- **Testing:** Google Test, QtTest, unit/integration testing, TDD
- **Databases:** PostgreSQL, SQLite, MySQL
- **Other:** C#/.NET 8, JavaScript, Java/Scala

Experience

C++ Software Engineer | [Innowise Group](#)

12/2024 - Present

Global software engineering and IT services company.

- Refactored a production C++17/Qt Widgets SIP call-flow into an MVC/MVVM-style architecture, decoupling UI state, user actions, media negotiation, and Linphone-SDK integration to simplify feature development and regression testing.
- Implemented resilient active/backup SIP registration and failover logic in Linphone-SDK/liblinphone, including registration watchdog recovery, realm-aware authentication, AuthInfo synchronization, and server-state persistence.
- Developed and hardened asynchronous SIP/WebRTC calling and video-conferencing flows, including multi-call handling, remote hangups, conference management, media switching, participant management, and audio/video controls.
- Implemented SNMP/DHCP/TFTP/SQLite device-provisioning workflows in a WinForms/.NET 8 utility, including TFTP delivery, DHCP option 66/67 flows, SIP configuration generation, HA1/MD5 authentication, and realm probing.
- Integrated WebRTC/mediasoup media services and V4L2/GStreamer camera/video stack on NXP i.MX Embedded Linux, including Yocto/BitBake customization, systemd configuration, and target-side deployment/debugging.
- Built QtTest unit/integration coverage for asynchronous SIP, media, and networking workflows, including headless/offscreen execution directly on the embedded target.
- Performed code reviews, contributed to task planning, and provided onboarding, technical mentoring, and interview support as the engineering team grew to 6 people.

Tech Stack: C++17, Qt 5.15.7/Qt Widgets, QtConcurrent/QThread, Embedded Linux, Linphone-SDK, SIP/WebRTC, mediasoup, Yocto/BitBake, NXP i.MX, systemd, QtTest, C#/.NET 8, SQLite, CMake, GCC/gdb.

Global digital engineering and IT services company.

- Developed and maintained C/C++ modules across two production enterprise systems: Energy ERP (C++17) and Digital Library (C89), and performed peer code reviews in distributed international engineering teams.
- Diagnosed production and integration issues on Linux systems hosted on AWS EC2 and Azure VMs using SSH, gdb, and core dump analysis.
- Digital Library: Maintained legacy C89 modules and resolved integration defects between the C core and Java/Scala services.
- Energy ERP: Optimized PostgreSQL schema and SQL queries, reducing search response time by ~80%, from 1.5s to 300ms.
- Maintained JavaScript integration modules connecting backend data flows with the Web UI.

Tech Stack: C89, C++14/17, PostgreSQL, Java 8/11, Scala, JavaScript, Linux, gdb, AWS/Azure, CMake, Jenkins.

C++ Software Engineer | RIFTEK**03/2020 - 05/2020**

Industrial metrology and 3D scanning software.

- Developed a 3D reconstruction R&D prototype using C++17, Qt, and Point Cloud Library (PCL).

Tech Stack: C++17, Qt, PCL.

MATLAB Algorithm Developer | PELENG**08/2012 - 01/2020**

Aerospace, Earth-observation and image-processing R&D.

- Developed Earth-observation and UAV image-processing algorithms, including feature-based image stitching, stabilization, debayering, and sensor fusion.
- Implemented custom lossy satellite-image compression, achieving a 3-4x compression ratio.
- Developed image-type prediction and quantization algorithms, improving image-quality metrics by ~30% and estimated Earth-observation accuracy by 15-20%.

Tech Stack: MATLAB, Git, MS Access.

Education

- BSc in Radiophysics - Satellite information systems and technologies, [Belarusian State University, Faculty of Radiophysics and Computer Technologies](#) | 2007 - 2012

Languages

English B2 · Polish B1 · Swedish A2 · Russian Native · Belarusian Native